BSES

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Climate Resilience and Securing the Physical Assets of DISCOMs

Session 5 | Day 2 | 11:45 – 13:00 IST

Why Climate Resilience Is Now a Core DISCOM Mandate

50°C+

Summer peak in Western India, 2024 – warmest year on record

270.82 GW

Record national peak in 2026 after four straight record days

1.3 million

Estimated DT failures a year from overloading and poor maintenance

4.5% of GDP

Estimated annual cost of extreme heat and humidity to India

Heat

- Conductor sag and equipment derating
- Higher DT failure rates
- All 50 of the world's hottest cities were in India in 2026

Cyclones

- Tens of thousands of poles and DTs flattened
- Overhead lines most exposed
- Lightning adds to the damage

Floods

- Substations inundated in Mumbai, Chennai, Delhi and other cities
- Urban flooding cuts essential services

Reliable power for cooling is a life-safety service, not a convenience

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Flood-Proof Siting and Faster Underground Cabling

1 Siting equipment in flood-prone areas

- Identify substations, RMUs, and DTs located in flood-prone zones.
- Use flood maps to assess risk and plan mitigation measures.
- Raise and flood-protect vulnerable substations.
- Provide proper waterproofing, drainage, and cable trench protection.
- Relocate critical equipment from low-lying areas where feasible.

2 Underground and covered conductors

- Use underground or covered conductors in cyclone-prone areas.
- Adopt faster installation methods and standardized materials.
- Coordinate early with local authorities for route approvals.
- Reduce outages caused by falling trees, strong winds, and lightning.

BSES in action: Kalkaji, Delhi

India's first regulator-approved 1000 KVA SF6-free underground digital smart substation, commissioned Dec 2025

Flood protection and drainage built into the civil design

80–85%

surface space saved

0.1–0.4%

annual O&M, vs 1.5–2% for conventional AIS

SCADA

remote monitoring

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Stronger Transformers and Data-Driven Risk Mapping

3 New DT design

- Implement the new DT design as per IS 1180.
- Prioritize deployment in high-failure and high-risk areas.
- Standardize specifications across DISCOMs for cost-effective procurement.
- Use load and temperature monitoring to detect overloading early.
- Maintain spare transformers for quick replacement.

4 Data-driven risk mapping and resilient design

1

Map

Map the network (GIS) against IMD hazard data for heatwaves, cyclones, floods, and lightning.

Identify vulnerable assets and high-risk areas for targeted action.

2

Assess

Prioritize feeders, DTs, and substations based on risk and criticality.

Identify assets most exposed to hazards and serving critical loads.

Focus investments on high-risk, high-impact locations.

3

Design

Build to stronger standards in cyclone and flood zones; rank capex by risk

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Disaster Preparedness and Rapid Restoration

1 Prepare

- Spares in stock for poles, DTs and cables
- Regular tree trimming along line corridors
- Mobile substations pre-positioned
- Mutual-aid agreements between DISCOMs

2 Respond

- Drones for quick damage assessment
- Self-healing feeders isolate faults automatically
- Priority restoration for hospitals and essential services
- Single control-room view of the event

3 Restore

- Deploy mobile substations
- Bring in crews from neighbouring DISCOMs
- Fast-track replacement of damaged assets
- Post-event review to update standards

Digital agility

Smart meters
outage and voltage data

ADMS + IoT
real-time network view

Automation
fault detection, predictive maintenance

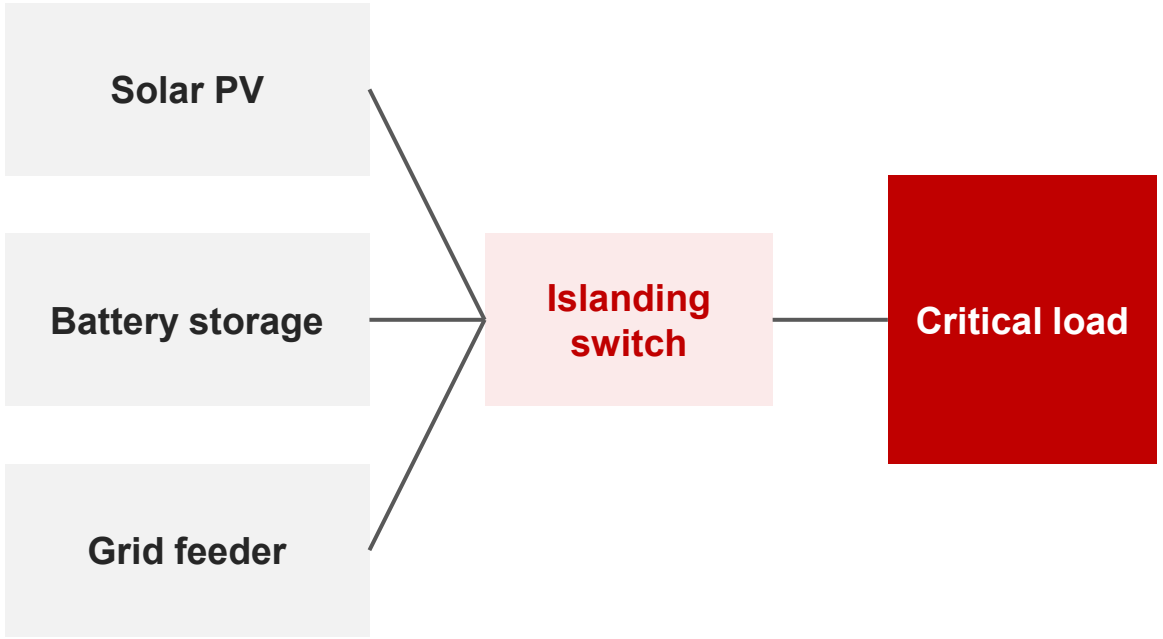
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Emergency Microgrids for Essential Services

Where and why

- Hospitals, water pumping, emergency services and shelters
- Decentralised solar plus battery that islands from the grid during an event
- Keeps life-safety loads running while the main network is restored
- Cuts transmission losses and supports the decarbonisation goal

How an emergency microgrid works



Normal operation: grid-connected. During an outage the switch isolates the microgrid and keeps essential loads on.

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Financing Grid Resilience and the Way Forward

Funding and Financing Avenues:

- Create a dedicated fund for quick emergency restoration.
- Recognize resilience investments in tariff approvals.
- Use insurance and pre-approved disaster response funds.
- Prioritize funding based on risk and vulnerability assessments.

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Thank You

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